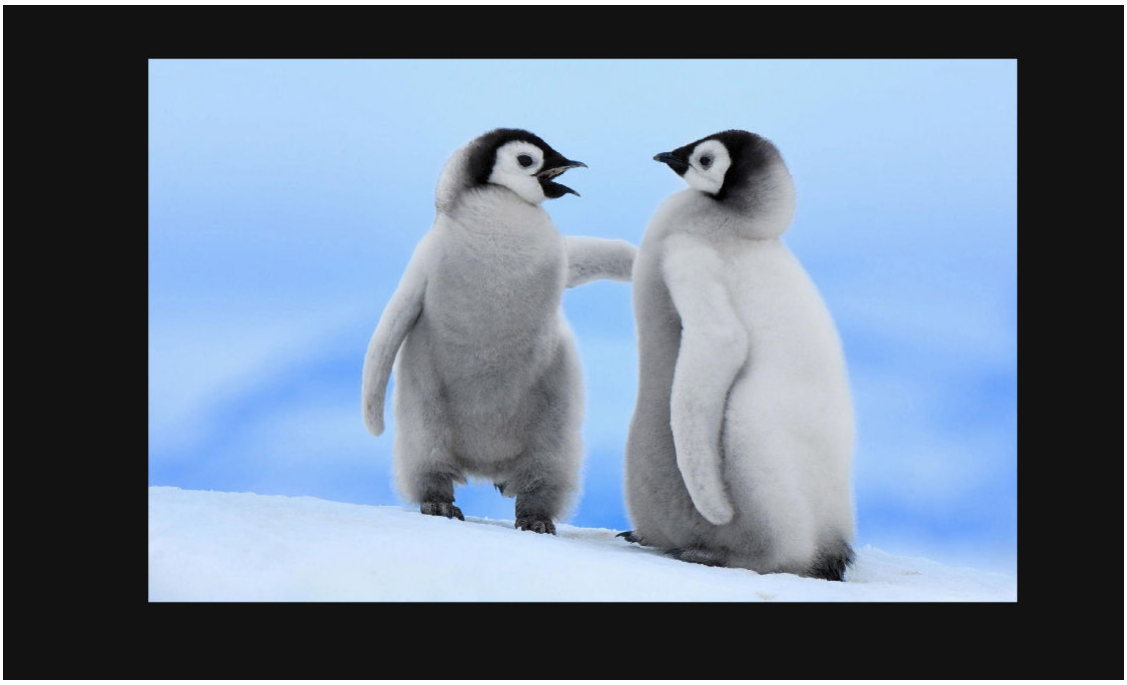
IMAGE PROCESSING

Color channels

```
I0 = imread("images/pengu.jpg");
imshow(I0)
```



```
size(I0)
```

```
ans = 1×3
        1800        2880           3
```

```
max(max(I0))
```

```
ans =
1×1×3 uint8 array
ans(:,:,1) =

   242


ans(:,:,2) =

   248


ans(:,:,3) =

   255
```

```
I = I0/255;
I_col = zeros(size(I));

for i = 1:3
    I_col(:, :, i) = I(:, :, i);
    figure
    imshow(I_col);
end
```



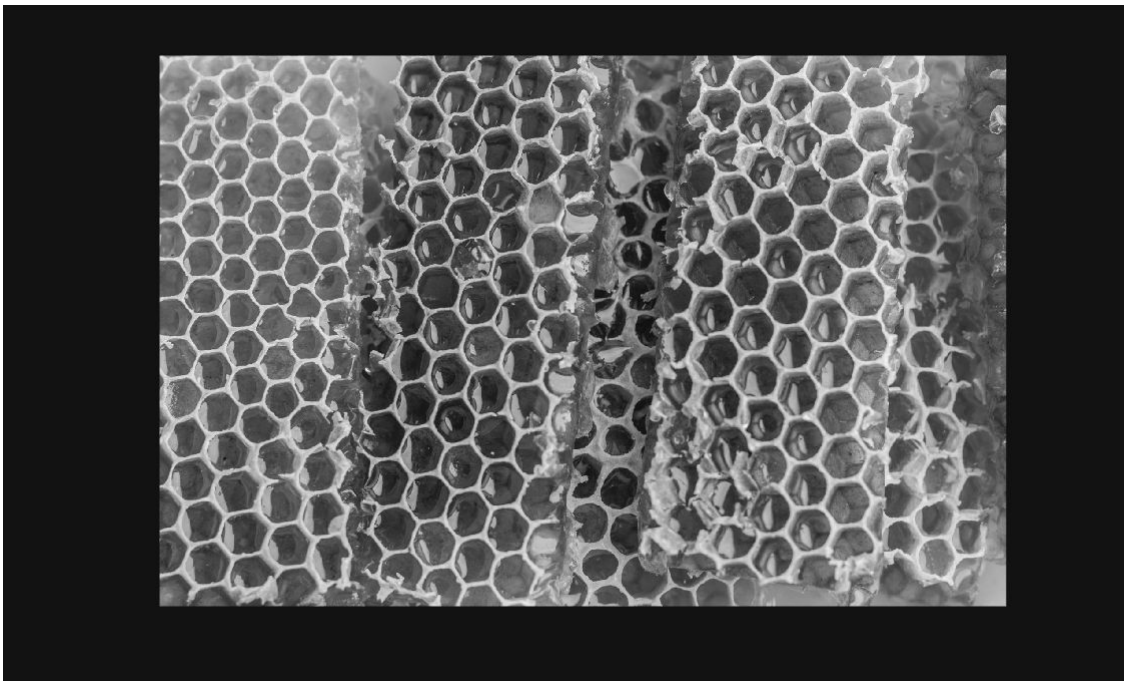




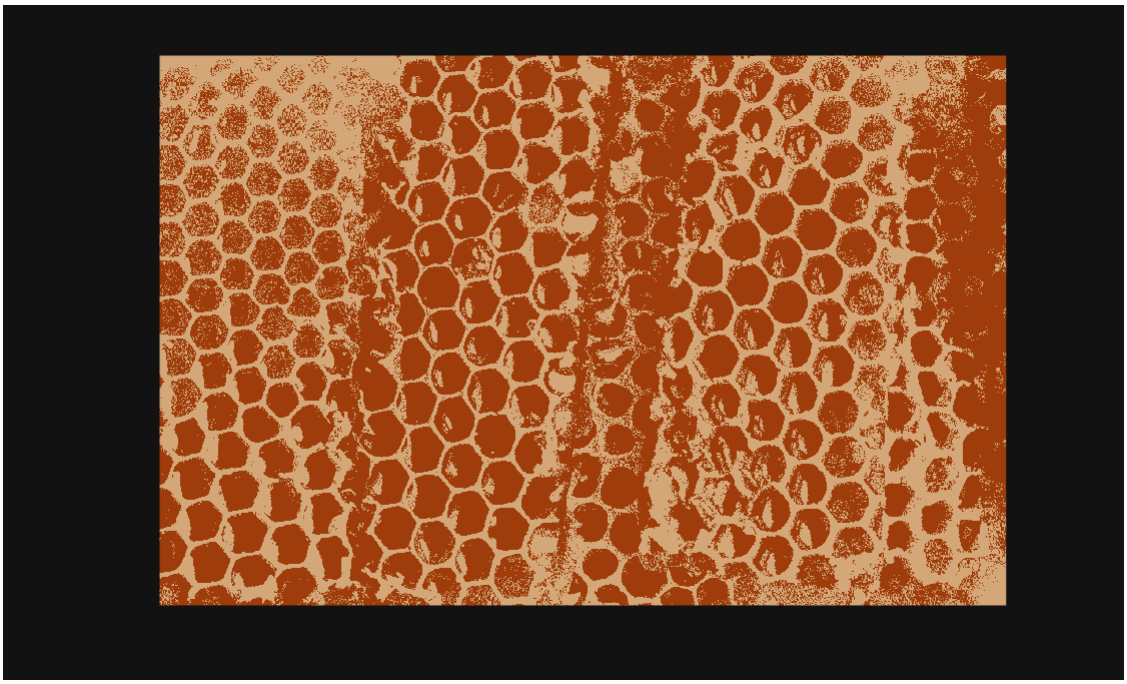
```
figure
```

Image Conversion

```
I1 = imread("images/honey-comb.jpg");
imshow(rgb2gray(I1))
```



```
for i = [2 5 50]
    [X, map] = rgb2ind(I1, i);
    figure; imshow(X, map)
end
```







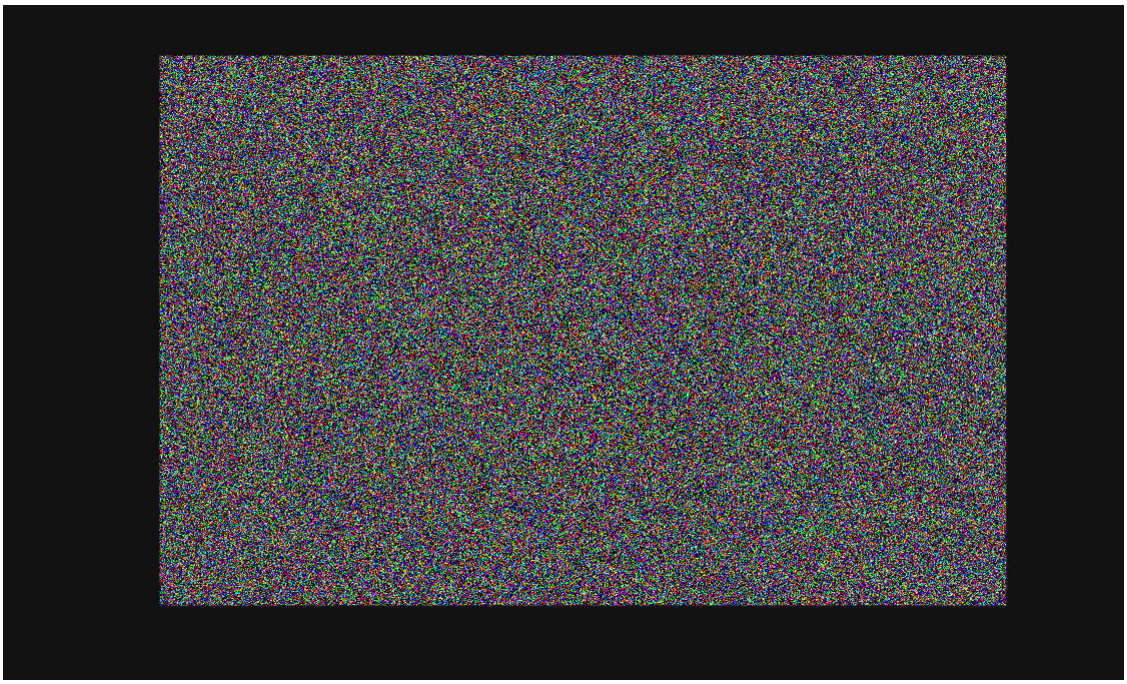
Sampling

```
[r, c, s] = size(I1);
N = 2;
I1s = I1/255;
y = zeros(size(I1s));
y(1:N:r, 1:N:c, :) = I1s(1:N:r, 1:N:c, :);
imshow(y)
```

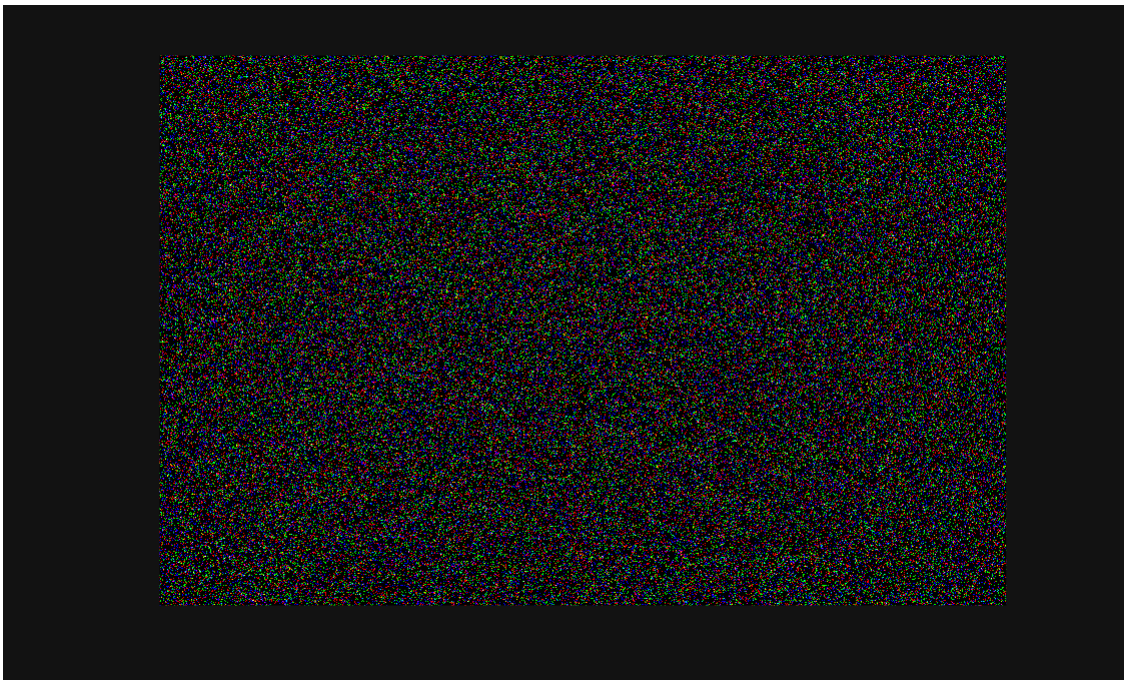


Noisy Image

```
imshow(imnoise(I1s, "gaussian", 0, 1));
```

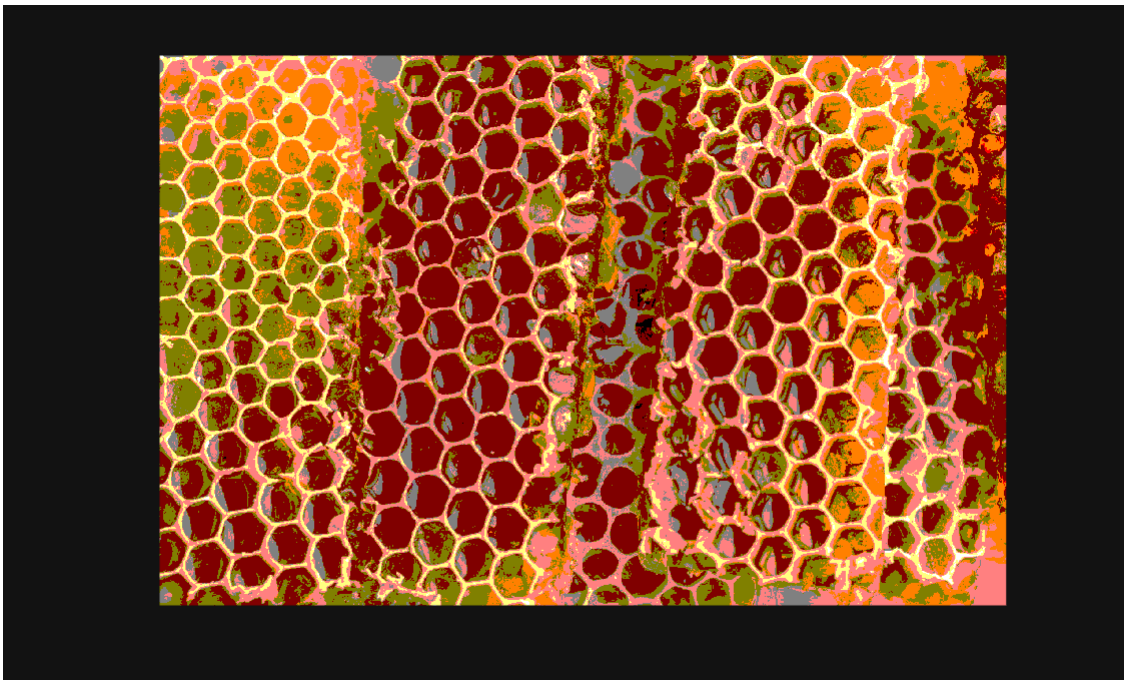


```
imshow(imnoise(I1s, "salt & pepper", 0.19));
```



Bit Resolution

```
b = [7 6 5];
for i = 1 : length(b)
    d = 2^b(i);
    z = round(I1/d);
    figure
    imshow(z*d)
end
```



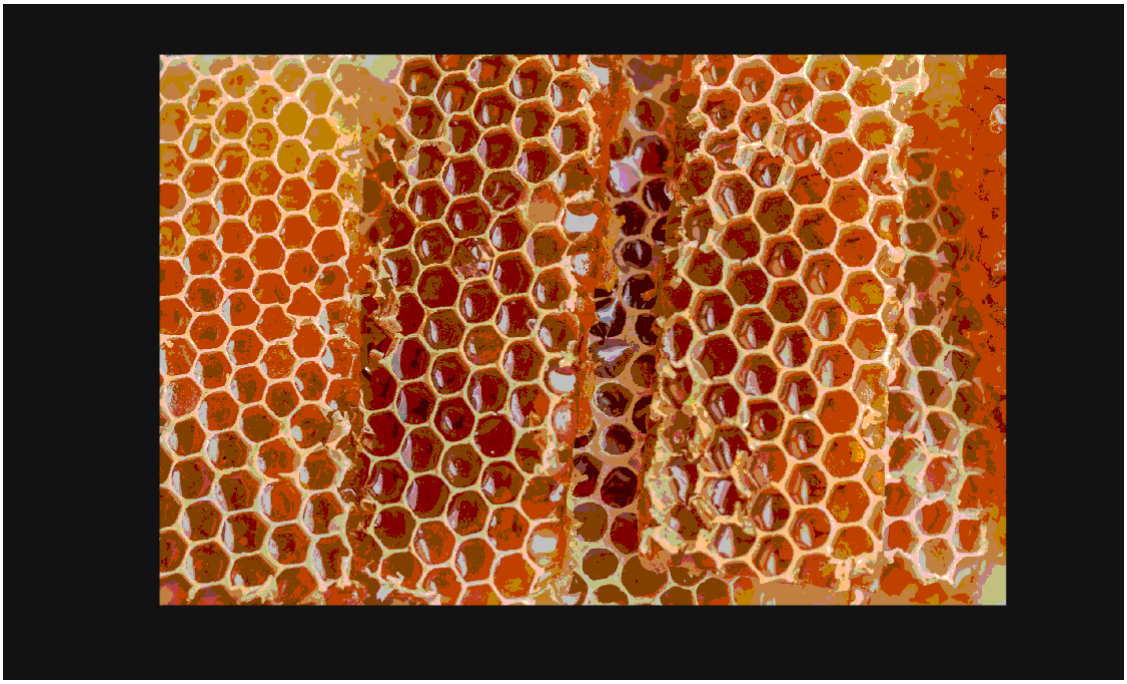




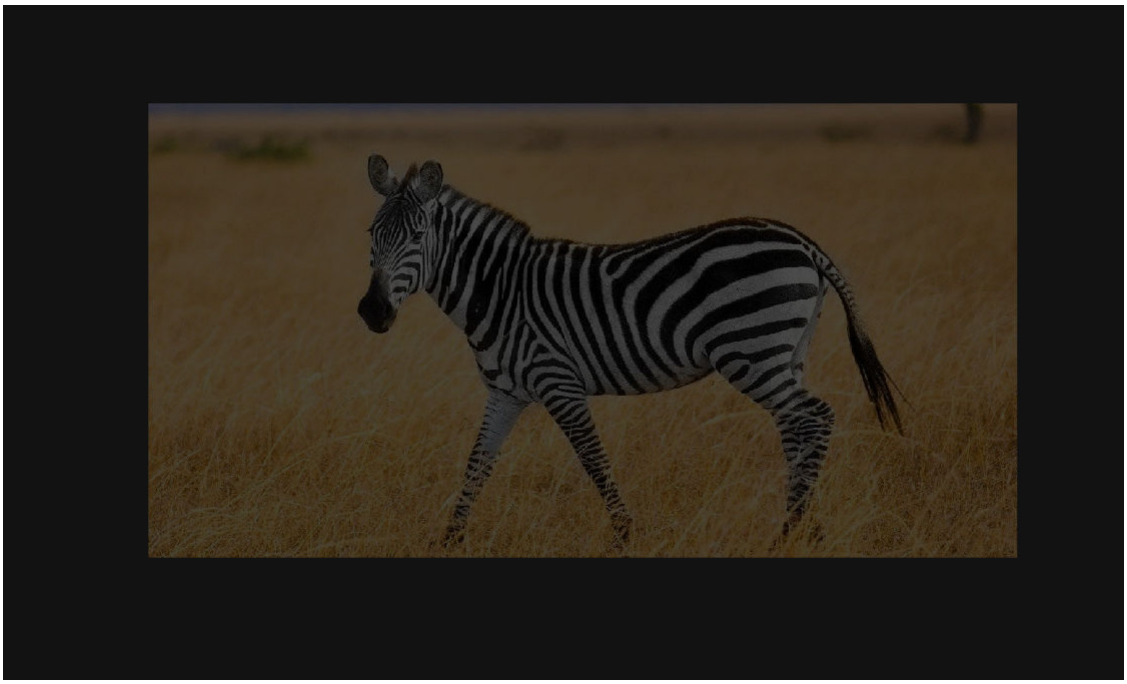
Image Operations

```
I2 = imread("images/zebra.jpg");
imshow(I2, []);
```



```
I1c = I1(1:627, 1:1200, :);

imshow(0.3 * I2);
```



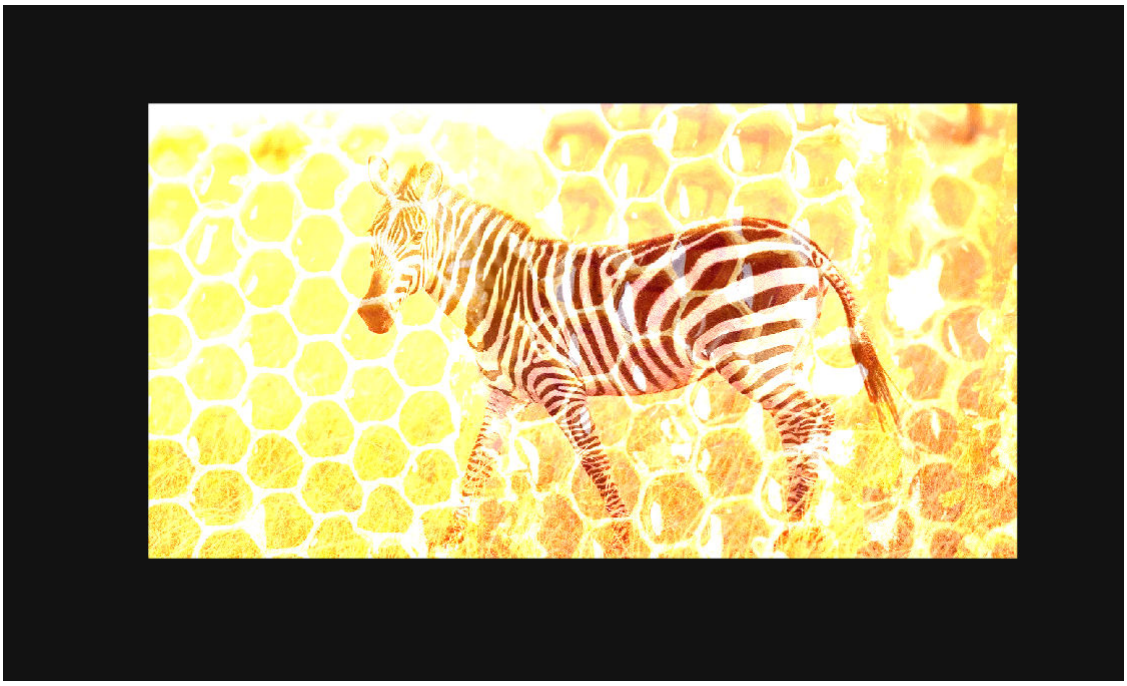
```
imshow(0.7 * I2 + 0.3 * I1c);
```



```
imshow(0.7 * I1c + 0.3 * I2);
```



```
% imadd()
imshow(imadd(I2, I1c));
```



```
imshow(imsubtract(I2, I1c));
```



```
imshow(I2 - I1c);
```



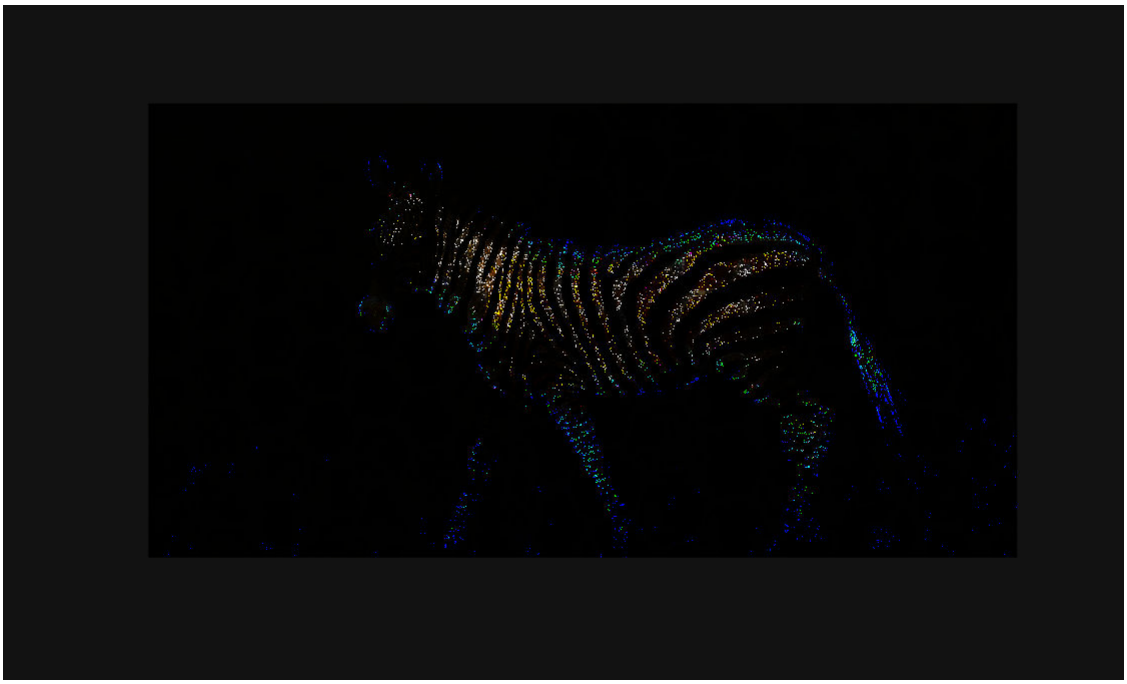
```
imshow(I1c - I2);
```



```
imshow(imabsdiff(I2, I1c));
```

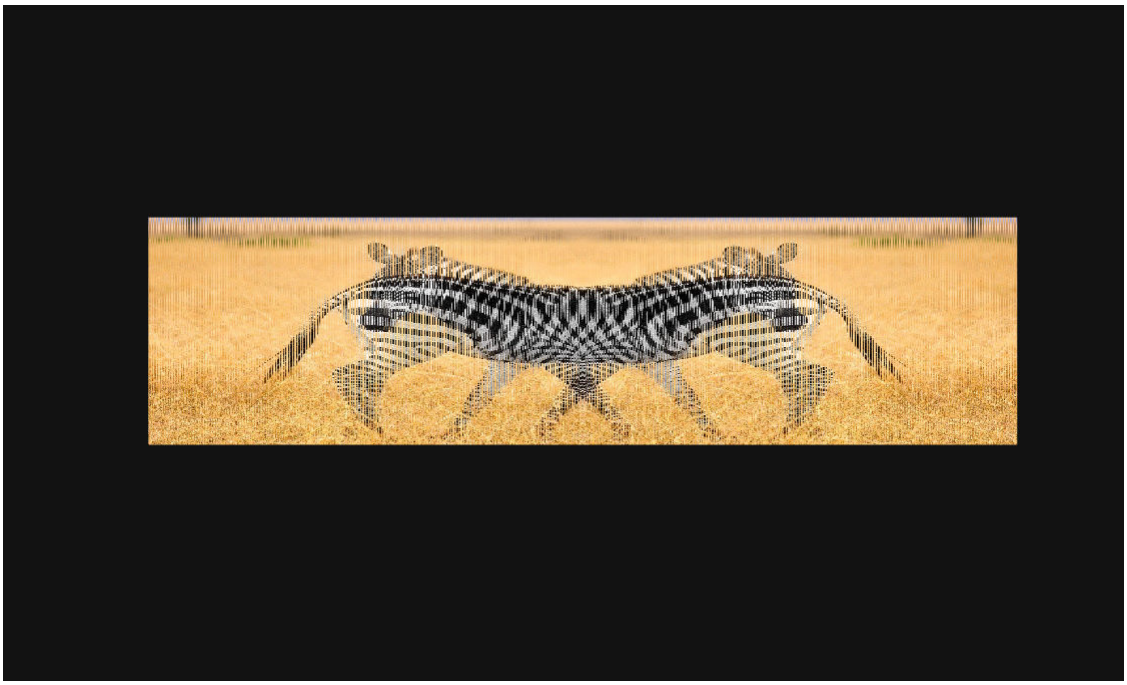


```
%image subtraction is useful for scene change detection
imshow(imdivide(I1c, I2));
```



Ex: Double the columns to have mirror image

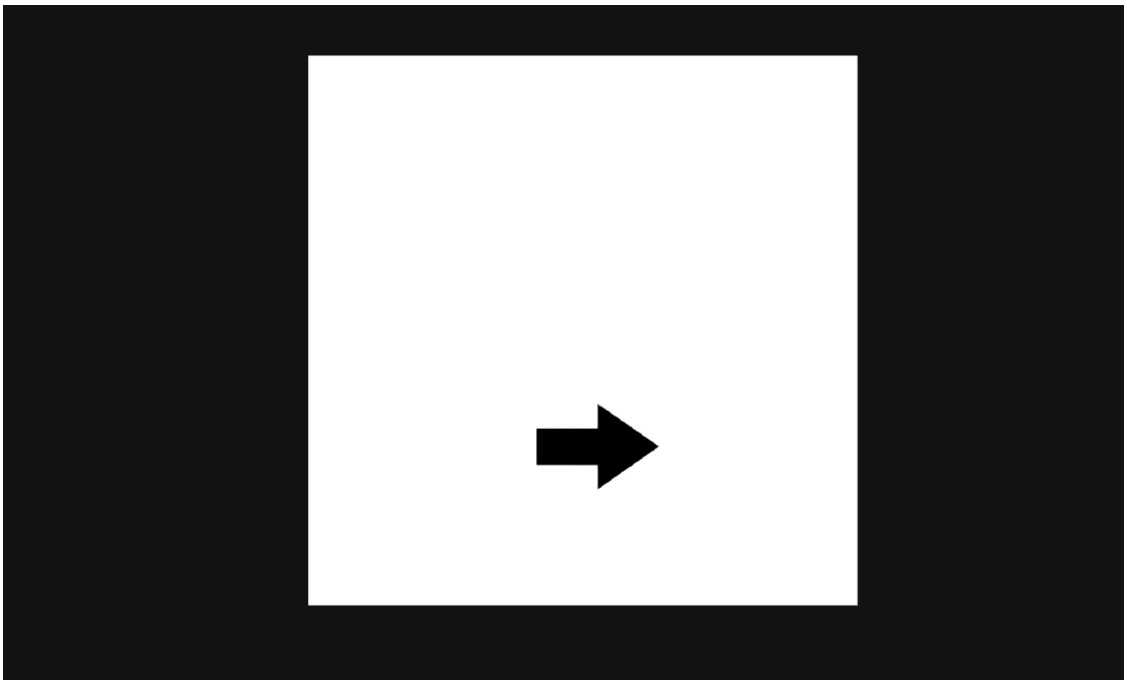
```
I2s = im2double(I2);
[r, c, s] = size(I2s);
res = zeros(r, c*2, s);
I2m = flip(I2s, 2);
res(:,1:2:size(res, 2)-1,:) = I2s(:,:,:);
res(:,2:2:size(res, 2),:) = I2m(:,:,:);
imshow(res);
```



Ex: Convulation vs Cross-correlation

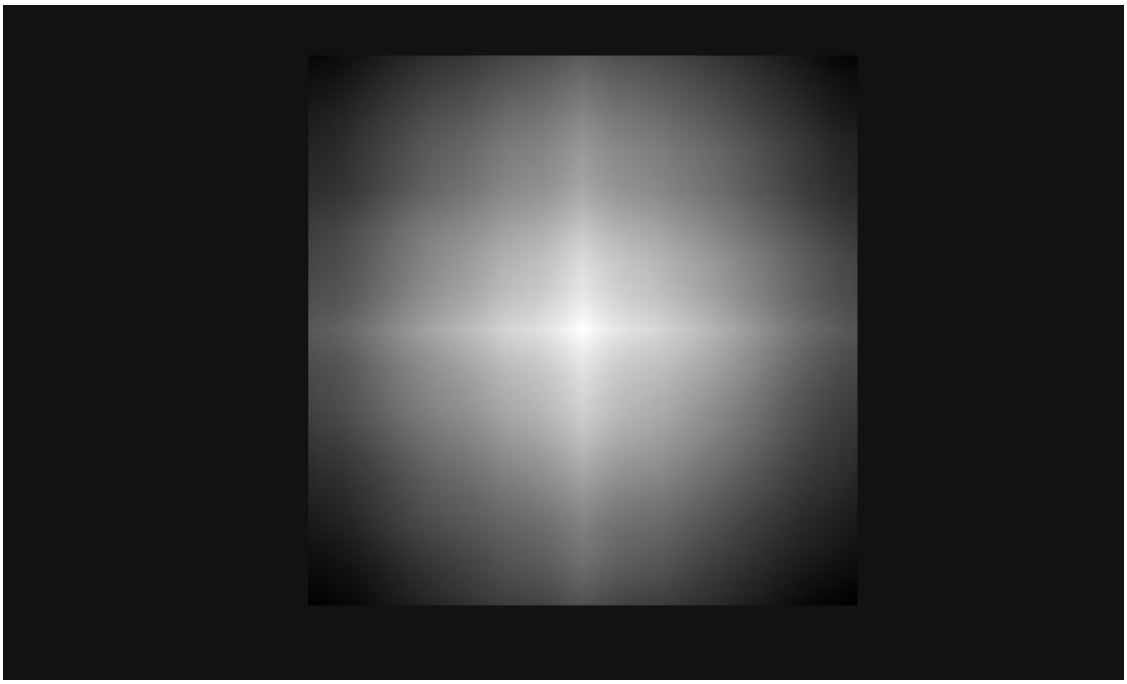
```
I3 = imread("images/arrows.jpg");
I4 = imread("images/arrow-right.png");
```

```
imshow(I3); imshow(I4);
```



```
img1 = double(rgb2gray(I3));
img2 = double(rgb2gray(I4));

C = conv2(img1, img2, 'same');
imshow(C,[])
```



```
figure

C_corr = conv2(img1, rot90(img2, 2), 'same');
imshow(C_corr,[]);
```

